

Pritam Dey

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Bio

I am a statistician working on developing statistical models motivated by real-world scientific problems. My current work spans methodological advances in variational inference and symbolic regression, as well as application-driven research in spatial transcriptomics, multi-omic data integration, neuroscience, and materials science. I aim to design efficient and robust models that enable and accelerate scientific discovery.

Work Experience

Postdoctoral Research Associate, Texas A&M University Sep 2023–Present

Working in several collaborative teams from the Departments of Statistics and Nutrition to develop Bayesian frameworks and computational tools for spatial transcriptomics, dietary microbiome studies, and multi-omic biomarker discovery.

Graduate Research Assistant, Duke University Jan 2019–Aug 2024

Research Assistant developing statistical models for brain structural connectomics, including outlier detection and hierarchical continuous network representation.

Graduate Teaching Assistant, Duke University Aug 2019–Aug 2024

Led labs and performed grading for graduate-level courses: Linear Models (Fall 2019), Probability (Summer 2020), Probability and Measure Theory (Fall 2020), Predictive Modelling and Statistical Learning (Fall 2021), Probabilistic Machine Learning (Spring 2023).

Education

Ph.D. in Statistical Science, Duke University 2018–2023

- Advised by Dr. David B. Dunson.
- **Dissertation Title:** Some Advances in Statistical modeling of Brain Structural Connectomes.

M.S. in Statistical Science, Duke University 2018–2023

Master of Statistics, Indian Statistical Institute, Kolkata 2016–2018

Bachelor of Mathematics (Hons.), Indian Statistical Institute, Bangalore 2013–2016

- **Award:** S.H. Aravind Gold Medal for academic performance

Technical Skills

- **Statistics:** Bayesian Statistics, Variational Inference, MCMC, High-dimensional data analysis, Tree-structured modeling, Stochastic processes.
- **Bioinformatics:** Spatial transcriptomics modeling, multi-omic integration, QIIME2, PICRUSt2, edgeR.
- **Programming:** Python (Pandas, NumPy, Scikit-learn), R (tidyverse), SLURM, MATLAB, C/C++, SQL.

Publications and Preprints

Published papers

2. **Outlier detection for multi-network data.** Dey, P., Zhang, Z., Dunson, D.B. *Bioinformatics* (2022).
1. **dame-flame: a Python package providing fast interpretable matching for causal inference.** Gupta, N.R., Orlandi, V., Chang, C.-R., Wang, T., Morucci, M., Dey, P., Howell, T.J., Sun, X., Ghosal, A., Roy, S., Rudin, C., Volfovsky, A. *Journal of Statistical Software* (2025).

Preprints and papers under review

5. **VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees.** Roy, S., Dey, P., Mallick, B.K. *arXiv preprint. Accepted at Uncertainty in Artificial Intelligence (UAI) 2026.*
4. **A Generalized Tangent Approximation Framework for Strongly Super-Gaussian Likelihoods.** Roy, S., Dey, P., Pati, D., Mallick, B.K. *arXiv preprint. Major revision (JASA, Theory and Methods)*
3. **JASPER: Joint Bayesian Analysis of Spatial Expression via Regression.** Dey, P., Guhaniyogi, R., Ni, Y., Mallick, B.K. *arXiv preprint. Submitted*
2. **Additive Nonparametric Regression with Spatial and Network Objects.** Guhaniyogi, R., Dey, P., Chandra, K., Scheffler, A., Mallick, B.K. *Major revision (Biometrics)*
1. **Hierarchical Bayesian Operator-induced Symbolic Regression Trees for Structural Learning of Scientific Expressions.** Roy, S., Dey, P., Pati, D., Mallick, B.K. *arXiv preprint. Submitted*

Manuscripts in preparation

1. **Joint Integrative Spatial Transcriptomics via Bayesian Modeling for Domain Recovery and Spatially Variable Gene Selection.** Dey, P., Guhaniyogi, R., Ni, Y., Mallick, B.K. *In preparation*
2. **Ensembles of Mondrian Processes for Continuous Modeling of Structural Connectomes.** Dey, P., Zhang, Z., Dunson, D.B. *In preparation*
3. **Hierarchical Ensembles of Mondrian Processes for Simultaneous Modeling of Common and Individual Structure of connectome Data.** Dey, P., Zhang, Z., Dunson, D.B. *In preparation*
4. **Noninvasive fecal multi-omic signatures for Apc^{S580/+}/Kras^{G12D/+} mice.** Ivanov, I., Mullens, D., Dey, P., Chung, H.C., Gaynanova, I., Ni, Y., Ufondu, A., Yang, F., Han, H., Woods, P., Goldsby, J.S., Davidson, L.A., Safe, S.H., Jayaraman, A., Chapkin, R.S. *In preparation*

5. **Diet shapes gut microbial function and indirectly influences host gene expression in infants.** Dey, P., Mullens, D., Ivanov, I., Chapkin, R., Donovan, S., et al. *In preparation*

Presentations & Conferences

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| 7. Contributed talk: JSM 2026, <i>Boston</i> | 2026 (<i>Scheduled</i>) |
| 6. Invited talk: SETCASA Annual Meeting, <i>Texas A&M University</i> | 2026 |
| 5. Invited talk: IISA Conference, <i>University of Nebraska–Lincoln</i> | 2025 |
| 4. Poster: TREC Annual Cancer Research Symposium, <i>Texas A&M University</i> | 2025 |
| 3. Contributed talk: NISS 2nd Annual Graduate Student Research Conference, <i>Online</i> | 2022 |
| 2. Contributed talk: WNAR Conference, <i>Online</i> | 2022 |
| 1. Poster: Statistical Methods in Imaging Conference, <i>Online</i> | 2021 |

Awards

3. **Rank 1** in National Eligibility Test (NET), conducted by Council of Scientific and Industrial Research (CSIR), India (2018).
2. **S.H. Aravind Gold Medal**, for outstanding performance in Bachelor of Mathematics (Hons.), ISI Bangalore (2016).
1. **KVPY Scholarship**, awarded by Department of Science and Technology, Govt. of India (2013–2018).